Sina Ronaghi

Metrologist | Data Analyst | Researcher
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LinkedIn | ResearchGate | GitHub | Online Portfolio

PROFILE SUMMARY

Multidisciplinary Electrical Engineer (Ph.D.) specialized in **Metrology**, **Data Science**, and **Signal Processing**. Proven track record within the high-tech industrial and R&D settings with a particular focus on entrepreneurship and project management. **Experienced leader** in startups and EU Flagship Programs with a hands-on and engaging attitude. Driven by curiosity and passionate about adopting **emerging AI technologies** for process optimization, task coordination, and improving team efficiency.

CORE COMPETENCIES

Signal Processing	Waveform Analysis, Harmonic Characterization and THD Analysis, Digital & Statistical Signal Processing, Sensor Fusion.
Engineering Tools	**Python, MATLAB & R** (Pattern Recognition, Predictive Maintenance, Machine Learning, Data Visualization & Signal Processing), **Databases** (SQL, MongoDB, InfluxDB), **Circuit Design & Analysis** (Altium, LTSpice), **Instrumentation & Automation** (NI LabVIEW/TestStand)
Metrology Framework	JCGM/GUM, CENELEC, WELMEC, EN, IEC, MID
Regulatory Exposure	EN 50160 (Voltage Characteristics), IEC 62052/53/56 (Smart Metering), IEC 61000 (EMC), ISO/IEC 17025 (Testing, Inspection) & Certification, ISO 9001 (QMS)
AI & Automation	Vertex AI, n8n, Node-RED, Ollama, Notebook LM, AWS, AZURE & GCP
Soft Skills	Certified LEAN Manager (Kaizen, 5S), Agile & DevOps (Trello, Jira & GitLab), Proactive, Public speaking, Cross-disciplinary stakeholder engagement

PROFESSIONAL EXPERIENCE

Post-Doc Researcher / Project Manager — **Nov 2025 – Present**

Politecnico di Milano — *Milan, Italy*

- Optimized R&D testing frameworks for VT/CT characterization (EN 50160) for Smart Grid Optimization via advanced signal processing in MATLAB/Python, increasing testing throughput by 70%.
- Directed ISO-compliant lab operations (IEC 17025 & ISO 9001), integrating LEAN management (Kaizen & 5S) to enhance workplace safety, and R&D output.
- Managed R&D proposal drafting for EU Flagship Programs (Erasmus+, Horizon) using Agile methodologies (Trello, Jira, GitLab) to synchronize milestones across +15 EMEA partners (e.g., Infenion, Imec, Siemens).
- Integrated AI-driven tools (Vertex AI, LLMs) into HW/SW workflows, automating R&D processes to accelerate development cycles.

R&D Associate Manager - Sustainable Mobility — **Nov 2024 – Oct 2025**

Politecnico di Milano — *Milan, Italy*

Recruited to lead a research project on electric vehicles (EV), while concurrently finalizing my Ph.D. dissertation.

- Led the design of high-precision sensor fusion systems for Sustainable Mobility (EV) applications.
- Automated test workflows by applying NI LabVIEW/TestStand, driving a 50% reduction in operational testing time compared to the legacy solution while ensuring measurement repeatability.
- Directed data-driven EV navigation predictive maintenance, employing unsupervised statistical learning and anomaly detection to ensure system reliability.

Ph.D. Researcher - Green Energy Integration & Smart Metering — **Nov 2021 – May 2025**

Politecnico di Milano — *Milan, Italy*

- Developed AI-driven predictive maintenance frameworks for smart metering compliance (IEC 62052/53/56)

via statistical learning, advanced signal processing, and anomaly detection with > 90% non-technical loss detection accuracy.
- Deployed scalable Cloud/IoT architectures (AWS/MQTT) for high-reliability data telemetry, ensuring 95%+ data integrity across 3+ years of continuous operation.
- Developed advanced signal processing algorithms for wearable health-tech, applying harmonic analysis to BioSignal (PPG & EEG) monitoring.
- Defined novel Signal Quality Assessment (SQA) indicators to enhance the reliability of wearable sensor data in health-tech.
- Formulated standardized metrological decision frameworks, contributing to international JCGM guidance for conformity assessment and risk-based decision frameworks.

Technical Consultant **Sep 2018 – Sep 2019**

Freelance contractor [Startup ecosystems] *Ahvaz, Iran*

- Deployed IIoT monitoring systems (M2M), optimizing ARM-based embedded hardware for EMC Compliance (IEEE 61000) in RF noisy environments for R&D purposes using Altium Designer & LTspice.
- Designed sensor fusion dashboards, leveraging software-driven control systems for real-time monitoring and data acquisition within the Agri-Tech sector via NI LabVIEW and Node-RED.

Co-Founder & CEO **Dec 2017 – Aug 2018**

Atiyeh Hoshyar LLC. [Startup] *Ahvaz, Iran*

- Co-founded an IoT startup and secured 60% of total funding through targeted pitches and incubator programs, accelerating early product development and market validation.
- Scaled engineering capacity by 40% in the first year by recruiting, onboarding, and leading a 5-person R&D team specializing in electronic design and embedded hardware/firmware.
- Filed a patent in Home Automation and Remote Monitoring, integrating secure M2M communication into ARM-based embedded HW/FW architectures.

TEACHING EXPERIENCE

Lecturer **A.Y. 2022 - A.Y. 2026 SEM I**

Digital Signal Processing (Advanced) (M.Sc.) *Politecnico di Milano, Milan, Italy*

- Co-delivered **196 hours** of lectures in Signal Processing, Data Acquisition Systems (DAQ), Time Synchronization & Positioning (GPS), and Instrument interfacing.
- Conducted live demonstrations, exam preparation sessions, student project evaluation, and written/oral examination rounds.

Teaching Assistant **A.Y. 2025/26 SEM I**

Measurement & Instrumentation (B.Sc.) *Politecnico di Milano, Milan, Italy*

- Delivered **56 hours** of hands-on lab lectures for fundamentals of electrical and electronic measurements, measurement systems for Industry 4.0, and practical demonstrations in NI LabVIEW, Python, and MATLAB.

EDUCATION

Ph.D. in Electrical Engineering (with honors) **Nov 2021 – May 2025**

Politecnico di Milano *Milan, Italy*

Thesis (public): Data-driven approaches for anomaly detection in energy metering.

M.Sc. in Electrical Engineering **Sep 2019 – Oct 2021**

Politecnico di Milano *Milan, Italy*

Thesis (public): Real-time motion capturing using inertial measurement units and applied embedded systems development

B.Sc. in Electrical Engineering **Sep 2013 – Jul 2018**

Shahid Chamran University *Ahvaz, Iran*

Thesis: Embedded Systems Design for Test Automation in AgriTech.

ADDITIONAL INFORMATION

Certificates	LEAN Management Fundamentals, Probability and Data with R, Python Data Structures, Data Scientist's ToolBox (Coursera); NI LabVIEW Object-Oriented Programming (Emerson); Signal and Power Integrity for High-Speed PCBs (Polimi); EU Driving License (B).
Awards	IEEE AMPS 2023 - Best Presentation Award; MIUR Scholorship for PhD Studies.
Speaking	Invited speaker at Milano Digital Week 2024, organized by the National Order of Engineers in Italy (CNI) for a 3-year regional energy monitoring project.
Committees	Technical Program Committee (TPC) for IEEE MetroXRAINE (since 2024).
Memberships	GMEE (Gruppo Misure Elettriche ed Elettroniche); IEEE Instrumentation & Measurement Society & IEEE Young Professionals (since 2021).
Languages	English (C2); Italian (B1).
Visa Sponsorship	Not required.

PUBLICATIONS

(J): Journal Publications (C): Conference proceedings

- (J) Risk-aware decisions: Taking into account admissible risk and measurement uncertainty in setting the acceptance limits - The European Physical Journal Conferences 2025
- (J) Obtaining a Probability Distribution From a Unimodal Possibility Distribution - IEEE Transactions on Instrumentation and Measurement 2024
- (J) A general Monte-Carlo approach to consider a maximum admissible risk in decision-making procedures based on measurement results - ACTA IMEKO 2023
- (J) Deconvolution-based methods to extract uncertainty components - Measurement Science and Technology 2023
- (J) A method to consider a maximum admissible risk in decision-making procedures based on measurement results - ACTA IMEKO 2023
- (J) Low-cost real-time motion capturing system using inertial measurement units - ACTA IMEKO 2022
- (C) Enhanced Resolution Analysis of Photoplethysmography Signals for Stress Monitoring Using Taylor-Fourier Method - IEEE EMBC 2025
- (C) Photoplethysmography Signal Quality Assessment Using Instantaneous Harmonic Analysis via Taylor-Fourier Method - IEEE I2MTC 2025
- (C) State of Charge Estimation of an Electric Battery using the Theory of Evidence - IEEE I2MTC 2025
- (C) Data-Driven Anomaly Detection for Smart Energy Meters - IEEE AMPS 2024
- (C) A Method to Obtain a Probability Distribution from a Unimodal Possibility Distribution - IEEE I2MTC 2024
- (C) Novel Algorithms for Filtering and Event Detection in Non-Intrusive Load Monitoring - IEEE AMPS 2024